
Creating a **better** **universe** with **diverse**

Mauro Lepore

Why > How > Who

Documenting a
collection of R packages
should not be hard.



The tidyverse is an opinionated
[collection of R packages](https://tidyverse.org/)

```
# Install from CRAN
install.packages("tidyverse")
```

```
library(tidyverse)
#> — Attaching core tidyverse packages — tidyverse 2
#> ✓ dplyr      1.1.3      ✓ readr      2.1.4
#> ✓ forcats    1.0.0      ✓ stringr   1.5.0
#> ✓ ggplot2    3.4.4      ✓ tibble    3.2.1
#> ✓ lubridate  1.9.3      ✓ tidyr     1.3.0
#> ✓ purrr      1.0.2
#> — Conflicts — tidyverse_co
#> ✗ dplyr::filter() masks stats::filter()
#> ✗ dplyr::lag()    masks stats::lag()
#> i Use the conflicted package (<http://conflicted.r-lib.org/>) to forc
```

All functions

`tidyverse_conflicts()`

Conflicts between the tidyverse and other packages

`tidyverse_deps()`

List all tidyverse dependencies

`tidyverse_logo()`

The tidyverse logo, using ASCII or Unicode characters

`tidyverse_packages()`

List all packages in the tidyverse

`tidyverse_sitrep()`

Get a situation report on the tidyverse

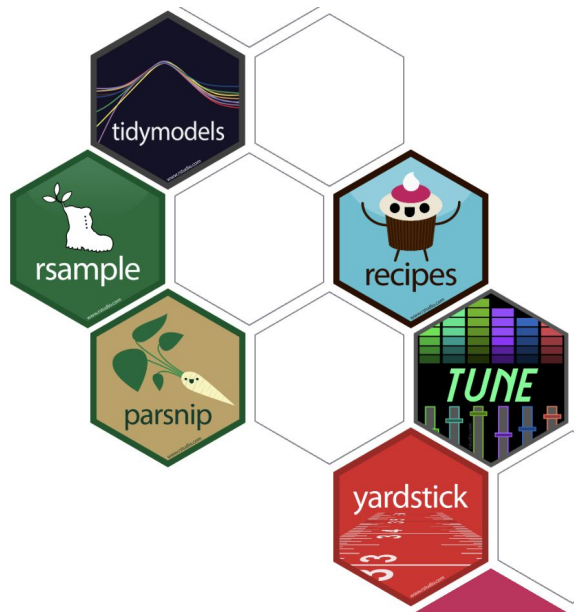
`tidyverse_update()`

Update tidyverse packages

As a user I want to:

Find all help files in a single place.

Ignore the individual packages.



Search all of tidymodels

Here are all the functions available across all of the tidymodels packages. Click on the link in the topic column to find the relevant reference documentation.

Show entries

Search: ×

title	topic	package
All	All	All
Adds the prediction columns to a piped command set.	tidypredict_to_column	tidypredict
Applies a calibration to a set of existing predictions	cal_apply	probably
Applies a calibration to a set of existing predictions	cal_apply.data.frame	probably

← Files

main ▾

tidymodels.org / find / all / index.qmd

↑ Top

Code

Blame



Raw



```
229
230   ```{r}
231   #| label: table-display
232   #| echo: false
233   tidymodels_functions %>%
234     datatable(rownames = FALSE,
235               class = 'cell-border stripe',
236               filter = 'top',
237               escape = FALSE,
238               options = list(pageLength = 25, autoWidth = TRUE))
239   ```
```

<https://github.com/tidymodels/tidymodels.org/blob/main/find/all/index.qmd>

As a developer I want to:

Have an easier way to create
a universe-wide reference.

Why > **How** > Who

dverse



document
your
universe

```
library(tidyverse)
```

```
packages <- tidyverse:::core
```

```
packages
```

```
#> [1] "ggplot2"    "tibble"     "tidyr"      "readr"      "purrr"      "dplyr"  
#> [7] "stringr"    "forcats"    "lubridate"
```

```
template <- "https://{package}.tidyverse.org/reference/{topic}.html"
```

```
docs <- dverse::document_universe(packages, template)
```

```
docs |> slice_sample(n = 5)
```

```
#> # A tibble: 5 × 7
```

```
#>   topic          alias title concept type keyword package  
#>   <chr>          <chr> <chr> <chr>   <chr> <chr>   <chr>  
#> 1 <a href=https://forcats.tidyverse.o... fct_c Conc... <NA>    help  <NA>    forcats  
#> 2 <a href=https://tibble.tidyverse.or... tibt... Pack... Datase... help  datase... tibble  
#> 3 <a href=https://lubridate.tidyverse... week... Get/... Data M... help  manip    lubrid...  
#> 4 <a href=https://lubridate.tidyverse... as_d... Conv... <NA>    help  <NA>    lubrid...  
#> 5 <a href=https://lubridate.tidyverse... peri... Crea... Data T... help  classes lubrid...
```

```
docs |> select(topic, alias)
#> # A tibble: 956 × 2
#>   topic                                                                 alias
#>   <chr>                                                                 <chr>
#> 1 <a href=https://forcats.tidyverse.org/reference/%>%.html>%>%</a>      %>%
#> 2 <a href=https://purrr.tidyverse.org/reference/%>%.html>%>%</a>        %>%
#> 3 <a href=https://stringr.tidyverse.org/reference/%>%.html>%>%</a>      %>%
#> 4 <a href=https://tidyr.tidyverse.org/reference/%>%.html>%>%</a>        %>%
#> 5 <a href=https://lubridate.tidyverse.org/reference/%m+%.html>%m+%</a>   %m+%...
#> 6 <a href=https://lubridate.tidyverse.org/reference/%within%.html>%withi... %wit...
#> 7 <a href=https://ggplot2.tidyverse.org/reference/+.gg.html>+.gg</a>     +, %...
#> 8 <a href=https://ggplot2.tidyverse.org/reference/.pt.html>.pt</a>       .pt,...
#> 9 <a href=https://ggplot2.tidyverse.org/reference/.pt.html>.pt</a>       .pt,...
#> 10 <a href=https://ggplot2.tidyverse.org/reference/CoordSf.html>CoordSf</... Coord...
#> # i 946 more rows
```

```
c("www.example.com", "Page not found") |> dverse::is_online()  
#> [1] TRUE FALSE
```

```
head(docs$topic) |> dverse::is_online()  
#> [1] FALSE FALSE FALSE FALSE FALSE FALSE
```

```
tail(docs$topic) |> dverse::is_online()  
#> [1] TRUE TRUE TRUE TRUE TRUE TRUE
```

```
docs |>
  filter(dverse::is_online(topic)) |>
  select(topic, title, package) |>
  DT::datatable(escape = FALSE)
```

Show entries

Search:

	topic	title	package
92	as_date	Convert an object to a date or date-time	lubridate
93	as_factor	Convert input to a factor	forcats
94	as_labeller	Coerce to labeller function	ggplot2
95	as_mapper	Convert an object into a mapper function	purrr
96	as_tibble	Coerce lists, matrices, and more to data frames	tibble
97	as_vector	Coerce a list to a vector	purrr

Why > How > **Who**

quarto publish cascadia.qmd



Quarto Pub

Published at <https://maurolepore.quarto.pub/cascadia/>



```
library(tidyverse)

template <- "https://{package}.tidyverse.org/reference/{topic}.html"
dverse::document_universe(tidyverse::core, template) |>
  filter(dverse::is_online(topic)) |>
  select(topic, title, package) |>
  DT::datatable(escape = FALSE)
```



Show 10 entries

Search:

	topic	title	package
92	as_date	Convert an object to a date or date-time	lubridate
93	as_factor	Convert input to a factor	forcats
94	as_labeller	Coerce to labeller function	ggplot2
95	as_mapper	Convert an object into a mapper function	purrr



billy



Global search

Source: [vignettes/articles](#)

Show ▾ entries

topic

2 [add_market_share](#)

57 [guess_numerical](#)

of a portfolio CSV

Global search

pacta.data.scraping
pacta.data.preparation
pacta.portfolio.import
pacta.data.validation
pacta.portfolio.allocate
pacta.portfolio.audit
pacta.portfolio.report
pacta.portfolio.utils
pacta.executive.summary
pacta.interactive.plot
pacta.scenario.data.preparation



Search:

package

pacta.scenario.data.preparation

pacta.portfolio.import

Showing 1 to 2 of 2 entries (filtered from 183 total entries)

Previous

1

Next





Show 10 entries

Search: adsl



	topic	title	package
3	admiral_adsl	Subject Level Analysis Dataset	admiral
7	adsl_peds	Subject Level Analysis Dataset-pediatrics	admiralpeds

Showing 1 to 2 of 2 entries (filtered from 11 total entries)

Previous

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Next

Functions

Show 10 entries

Search: adsl

	topic	title	package
5	admiralvaccine_adsl	Subject Level Analysis Dataset - Vaccine Specific	admiralvaccine
153	derive_vars_period	Add Subperiod, Period, or Phase Variables to ADSL	admiral

Showing 1 to 2 of 2 entries (filtered from 237 total entries)

Previous

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Next

On this page

Vignettes

Datasets

Functions



```
# _pkgdown.yml
```

```
navbar:
```

```
  type: default
```

```
  left:
```

```
    - text: "Reference"
```

```
      href: articles/reference.html
```





Creating a better universe with dverse

How to create a global search across an entire collection of R packages: A use case for the toy 'admiralverse'

TECHNICAL



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<https://bit.ly/dverse-pharmaverse>



Takeaways

Documenting a universe used to be hard.

You can now do it with `dverse::document_universe()`

It's great for universe developers but accessible to everyone.

Thanks



Mauro Lepore ✓ He/Him

[in/mauro-lepore](https://www.linkedin.com/in/mauro-lepore)

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